PROJECT REQUIREMENTS & COSTS

AI-Powered Employee Attrition & Workforce Planning

Presentation Document

1. Project Overview

The project analyzes employee data, attrition patterns, recruitment expenses, and workforce requirements. It uses data analytics to help organizations understand employee turnover and plan hiring more effectively.

2. Project Requirements

Hardware Requirements

- Laptop or desktop computer
- Recommended: 8 GB RAM or more
- Internet connection for software downloads and learning resources

Software Requirements

Tool	Purpose
Excel / LibreOffice Calc	Data inspection, cleaning, PivotTables, and charts
Python	Data analysis and machine-learning practice
MySQL	Store, query, filter, and join datasets
Power BI Desktop	Build interactive dashboards and reports

Data Requirements

- Employee details, department, role, and employment status
- Employee satisfaction and performance information
- Recruitment pipeline and hiring records
- Recruitment, onboarding, and training costs
- Current headcount, target headcount, and staffing needs

3. Estimated Project Costs

The project can be completed at low cost by using free tools and an existing computer. The table below estimates additional software and dataset costs for a student project.

Item	Estimated Cost	Notes
Laptop / computer	₹0 additional	Assumes a computer is already available
Excel alternative (LibreOffice Calc)	₹0	Free spreadsheet software
Python	₹0	Free to install and use
MySQL Community Edition	₹0	Free database software
Power BI Desktop	₹0	Free desktop authoring tool; some sharing features may require paid licensing
Synthetic practice dataset	₹0	Created for learning and demonstration
Total estimated additional cost	**₹0**	**Assumes existing hardware and no paid services**

Note: These are illustrative estimates, not a vendor quotation. Actual costs may vary if a new computer, paid software, cloud hosting, or additional services are required.

4. Expected Benefits

- Identify attrition patterns across departments.
- Understand recruitment and training expenses.
- Highlight possible workforce shortages.
- Support more informed recruitment and retention planning.
- Present findings through clear reports and dashboards.

5. Conclusion

This project combines Excel, SQL, Python, and Power BI to study employee attrition and workforce planning. Using free tools and synthetic practice data helps keep the learning project affordable. Any business conclusions should be based on verified analysis and validated data.

Presentation Reminder

The employee dataset used for practice is synthetic, not real company data. Clearly describe costs as estimates and avoid presenting illustrative figures as actual expenses.